FLUX

Molecules. Microbes. Outbreaks. One Platform.

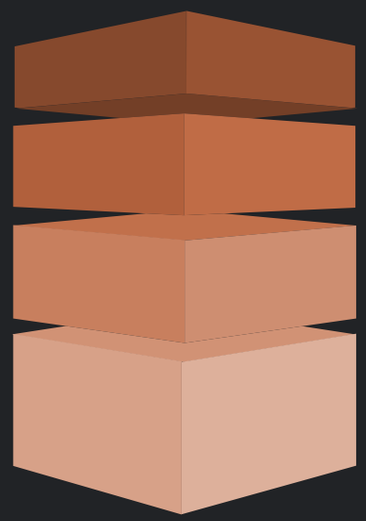
The Unified Intelligence Matrix for Global Health Security

Foundation

Integrate molecules and microbial data streams.

Analysis

Apply AI to detect outbreaks and patterns.



Aggregation

Unify surveillance, lab and field inputs.

Response

Coordinate interventions across stakeholders.

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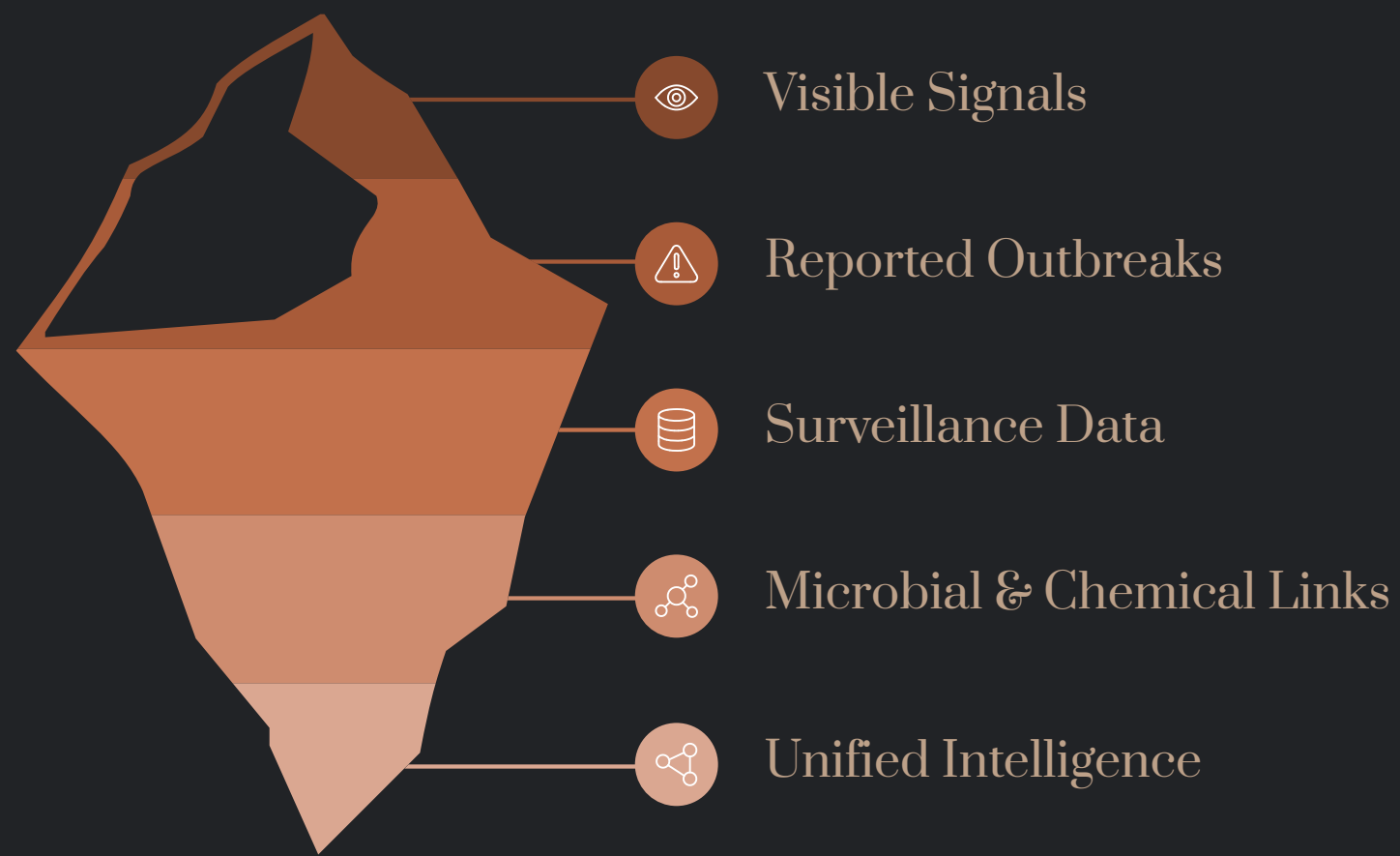
Health Risks Are Connected, But Our Data Is Fragmented

Today's health security landscape operates in isolated silos. Chemical safety, pathogen resistance, and epidemic spread are treated as separate challenges. Flux bridges these gaps with unified intelligence.



Safety Layer

Chemical toxicity prediction using molecular fingerprints and graph neural networks

Resistance Layer

Antimicrobial resistance patterns from sparse clinical datasets

Spread Layer

Epidemiological forecasting with vaccination and behavioral signals

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Engineering for Reality, Not Just Benchmarks

Dimension	Field Standard	Flux Approach
Scope	Single-track models	Molecule-to-population
Validation	80/20 random split	10-fold stratified CV
Deployment	Local notebooks	Live cloud dashboard
Ethics	Ignore imbalance	Masked focal loss + SMOTE-NC

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Molecular Toxicity: Scaling Safety with Hybrid GNNs

Deep Chemical Intelligence

MACCS keys encode molecular fingerprints as binary vectors

Graph neural networks capture topological patterns in chemical structures

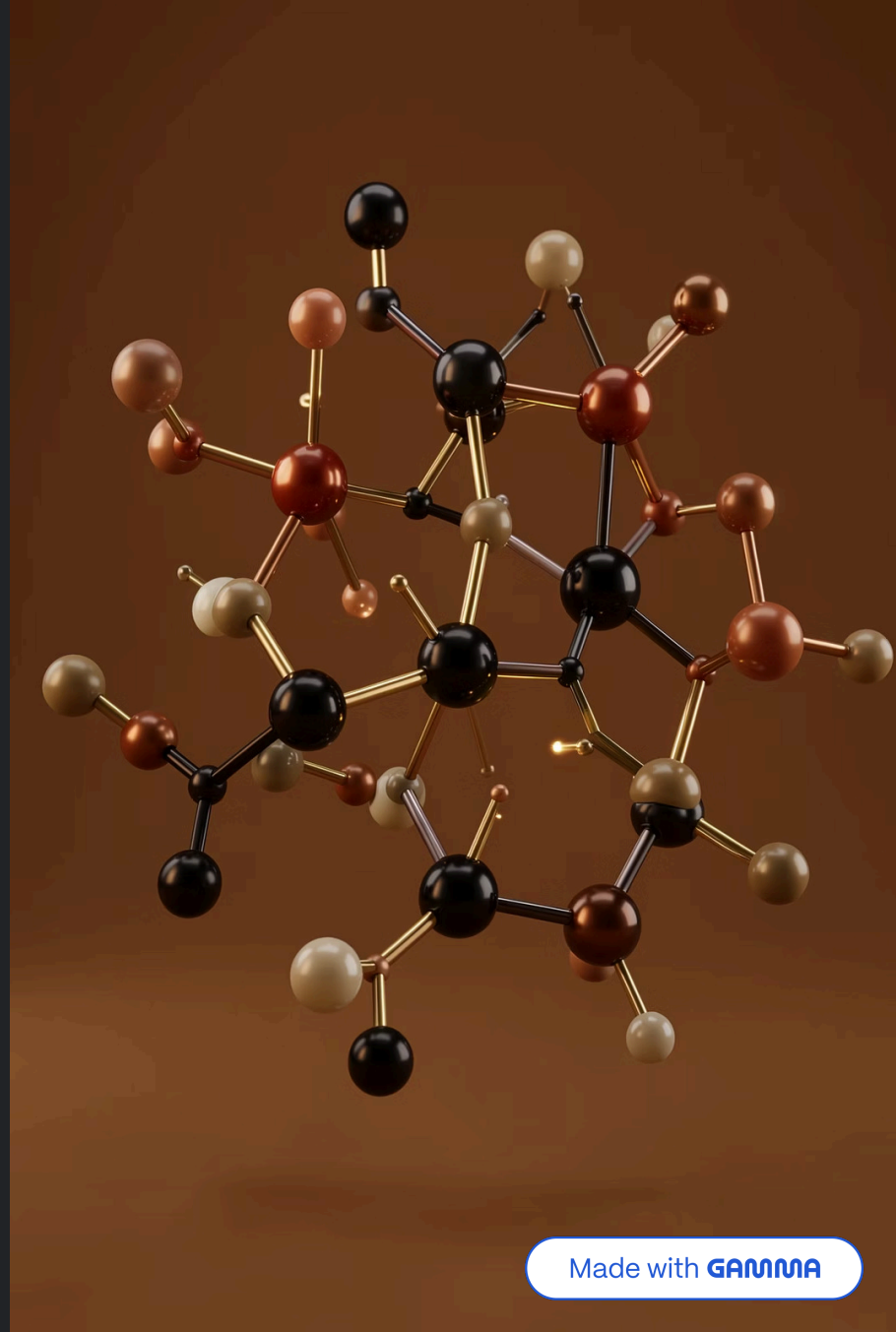
Knowledge distillation compresses heavy Teacher models into lightweight Students for sub-second inference

Tabular fusion combines GNN embeddings with handcrafted features

Performance

0.8416

AUC Score



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Antibiotic Resistance: Trustworthy Modeling on Sparse Clinical Data

10-Fold Stratified CV

Each fold validated on unseen data

Fold-Isolated SMOTE-NC

No data leakage across validation splits

Chronological Splits

Predicting actual future patterns

Flux's **mathematical honesty** ensures predictions reflect real-world conditions, not benchmark artifacts. Training on 274 clinical isolates with fold-isolated SMOTE-NC prevents data leakage.

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Epidemic Prediction: Real-World Signals, Real-Time Forecasts

Spatiotemporal Intelligence

LightGBM forecasting with lagged features

OWID integration for vaccination rates

Behavioral signals from testing data

Chronological splits prevent time-travel bias

0.8751

R^2 Score

Explained variance

10

Fold CV

Stratified validation

274

Clinical Isolates

Resistance patterns

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Flux is Deployed. Flux is Live.

A cloud-ready Streamlit dashboard connects all three tracks. Unified navigation, multi-track evaluation, and evaluator-friendly workflows enable real-time decision-making.

Unified Navigation

Single pane of glass for molecular, pathogen, and population layers

Multi-Track Evaluation

Compare model performance across all three tracks simultaneously

Evaluator-Friendly Workflow

Pre-configured test sets, validation protocols, and result exporters

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From Molecule to Outbreak

01

Scalable Architecture

Cloud-deployed platform, not isolated notebooks

02

Rigorous Validation

Stratified CV, temporal splits, fold-isolated augmentation

03

Unified Intelligence

One dashboard for toxicity, resistance, and spread

> **Trustworthy health AI starts with Flux.**

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